



speedup vs NumPy (right of the line = faster than NumPy; log scale)

NPBench PolyBench-derived kernels
preset paper, REPEAT=50 - VERIFICATION RUN, not a performance measurement
DaCe column = dace_cpu_autoopt / b27aaed2f
↓ = faster than NumPy, ↑ = slower; only validated results carry a speedup

kernel	NumPy (baseline)	Pluto	DaCe CPU	DaCe variant	notes
arc_distance		missing	invalid		dace:
azimint_hist		missing	invalid		dace:
azimint_naive		missing	invalid		dace:
cavity_flow		missing	invalid		dace:
channel_flow		missing	invalid		dace:
cholesky2		missing	invalid		dace:
compute		missing	invalid		dace:
contour_integral		missing	invalid		dace:
conv2d_bias		missing	invalid		dace:
covariance		missing	invalid		dace:
covariance2		missing	invalid		dace:
crc16		missing	invalid		dace:
doitgen		missing	invalid		dace:
gesummv		missing	invalid		dace:
go_fast		missing	invalid		dace:
hdiff		missing	invalid		dace:
jacobi_1d		missing	invalid		dace:
k2mm		missing	invalid		dace:
k3mm		missing	invalid		dace:
lenet		missing	invalid		dace:
mandelbrot1		missing	invalid		dace:
mandelbrot2		missing	invalid		dace:
mlp		missing	invalid		dace:
nbody		missing	invalid		dace:
resnet		missing	invalid		dace:
scattering_self_energies		missing	invalid		dace:
softmax		missing	invalid		dace:
spmv		missing	invalid		dace:
stockham_fft		missing	invalid		dace:
symm		missing	invalid		dace:
vadv		missing	invalid		dace: